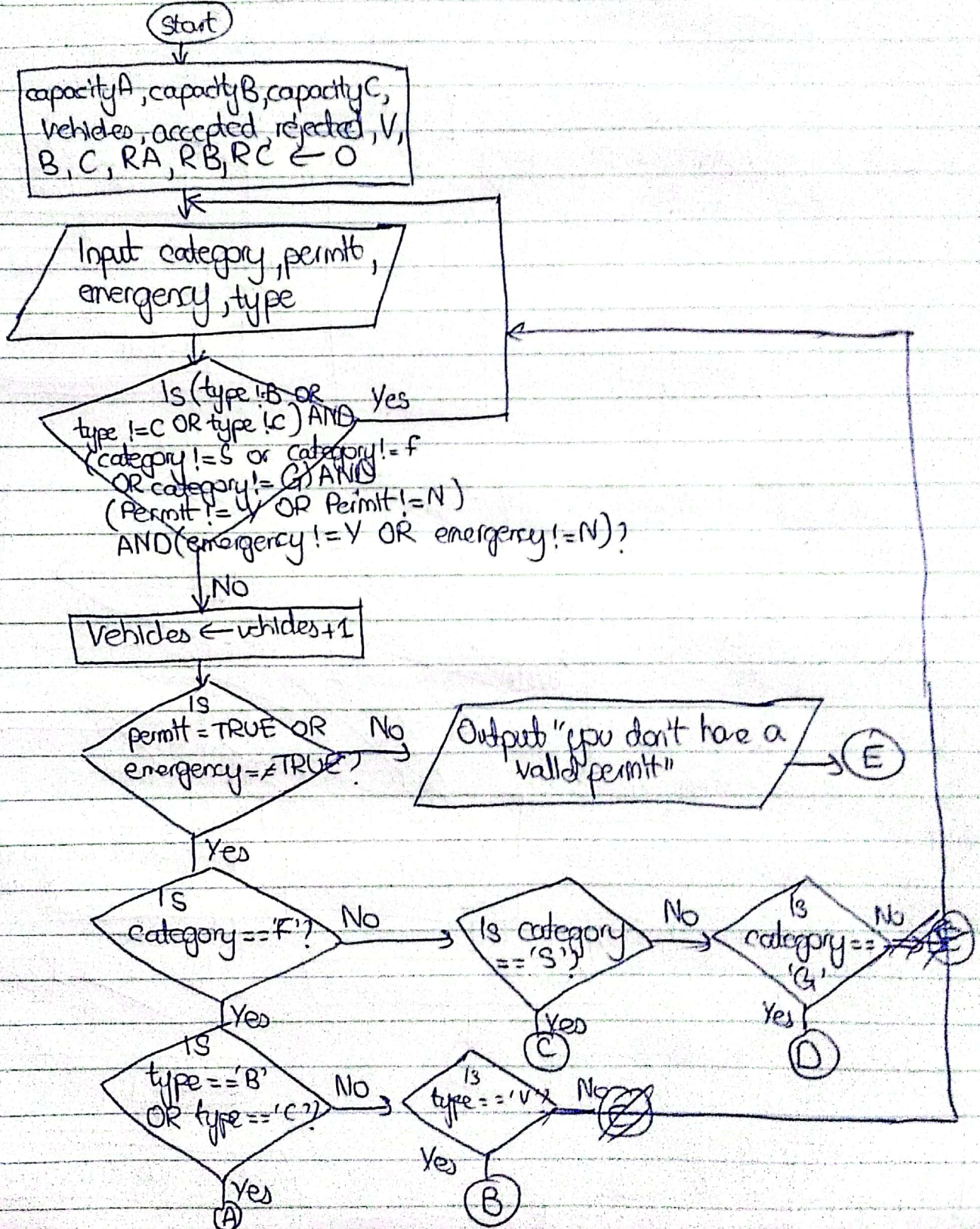
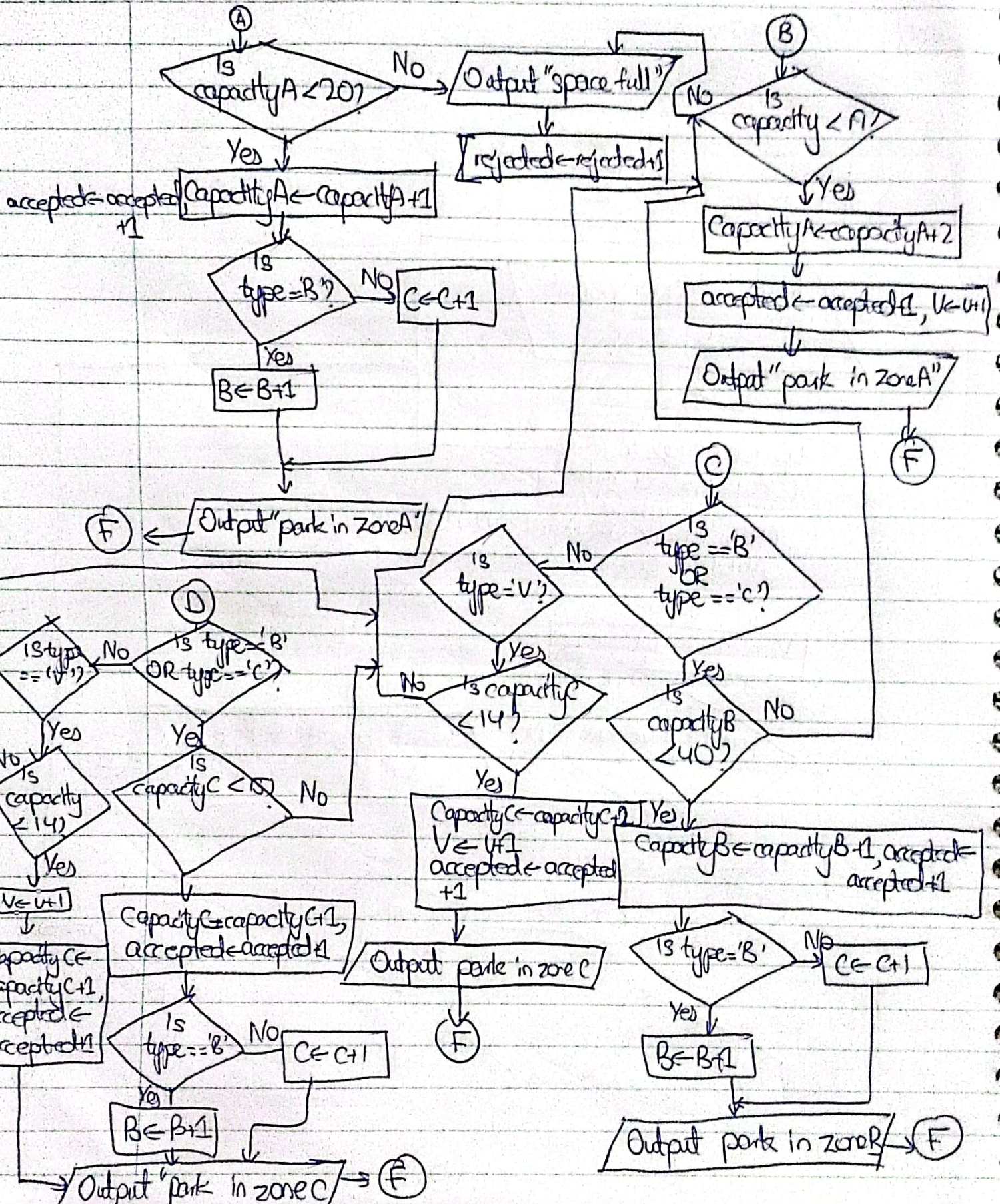
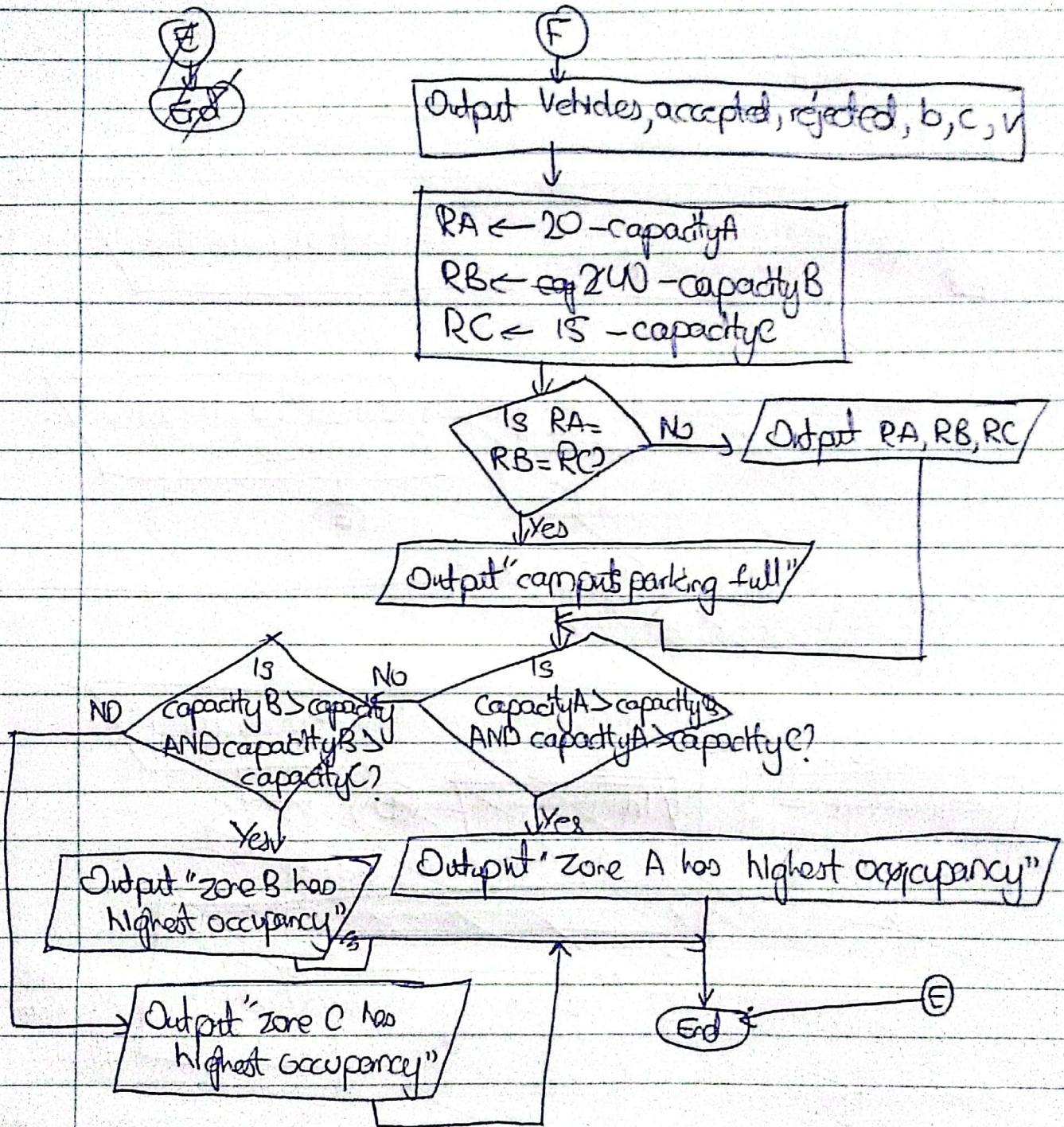


Q8.

Flowchart:







Date: _____

vehicles, accepted, rejected, B, V, C, RA, RB, RC

Q5. Declare capacityA, capacityB, capacityC : INT

Declare type: Char

Declare category: char

Declare permit, emergency : Boolean

CapacityA, capacityB, capacityC, vehicles, accepted, rejected, B, V, C, RA, RB, RC = 0

Output "Enter type (B, C, V), category (F, S, G), Permit (Y/N), emergency (Y/N): "

Input type, category, Permit, emergency

while (type != B AND type != C AND type != V) AND (category != F OR category != S OR category != G) AND (Permit != Y OR Permit != N) AND (emergency != Y OR emergency != N)

Do

Output "Invalid data, enter again:"

Input type, category, Permit, emergency

Endwhile ← vehicles ← vehicles + 1

If Permit = TRUE OR emergency = TRUE then

If category == 'F' then

If type == 'B' OR type == 'C' then

If capacityA < 20 then

If type == 'B' then → capacityA ← capacityA + 1

B ← B + 1

Else

C ← C + 1

EndIf

Output "park in zone A" accepted ← accepted + 1

Output "Space full", rejected ← rejected + 1

Else if type == 'V' then

If capacityA < 19 then

capacityA ← capacityA + 2

Output "park in zone A", accepted ← accepted + 1

Else

Output "space full", rejected ← rejected + 1

EndIf

V = V + 1 →

```

EndIf
Else If EndIf category == 'S' then
    If type == 'B' OR type == 'C' then
        If capacityB < 40 then
            capacityB ← capacityB + 1
            Output "you may park in zone B", Accepted ← accepted + 1
        Else
            Output "space full", Rejected ← rejected + 1
            If type == 'B' then
                B ← B + 1
            Else
                C ← C + 1
            EndIf
        EndIf
    EndIf
Else If type == 'V' then
    If capacityC < 14 then
        capacityC ← capacityC + 2, V ← V + 1
        Output "Park in Zone C", Accepted ← Accepted + 1
    Else
        Output "space full", Rejected ← rejected + 1
    EndIf
EndIf
EndIf

Else if category == 'G' then
    If type == 'B' OR type == 'C' then
        If capacityC < 15 then
            capacityC ← capacityC + 1, Accepted ← accepted + 1
            Output "Park in Zone C"
        Else
            Output "space full", Rejected ← rejected + 1
        EndIf
    EndIf
    If type == 'B' then
        B ← B + 1
    Else
        C ← C + 1
    EndIf
EndIf

Else if type == 'V' then
    If capacity < 14 then
        V ← V + 1
    EndIf
EndIf
    
```

```

    capacityC ← capacityC + 2
    Output "Park in Zone C", Aaccepted ← accepted + 1
Else
    Output "Space full", Rrejected ← rejected + 1
EndIf
EndIf
EndIf
EndIf
Else
    Output "No valid permit"
EndIf
Output "Total Vehicles are", vehicles, "Out of which", accepted,
"were accepted and", rejected "were rejected". Out of those
accepted vehicles, b, "were bikes", c, "were cars" and, v,
"were vans."
RA = 20 - capacityA
RB = 40 - capacityB
RC = 15 - capacityC
If (RA = RB = RC = 0) then
    Output "Campus parking facility is full"
Else
    Output "Capacity for zone A: ", RA, "Zone B: ", RB,
    "Zone C: ", RC
EndIf
If CapacityA > capacityB and capacityA > capacityC then
    Output "zone A has highest occupancy"

```



Date: _____

Else if capacity B $>$ capacity C AND capacity B $>$ capacity A then
Output "Zone B has the highest occupancy"

Else

Output "Zone C has highest occupancy"

EndIf

EndIf

→ Algorithm:

Else
EndIf

Output "you don't have a valid permit"